

Appendix D

Run-time errors

Applications generated by Free Pascal might generate run-time errors when certain abnormal conditions are detected in the application. This appendix lists the possible run-time errors and gives information on why they might be produced.

1 Invalid function number

An invalid operating system call was attempted.

2 File not found

Reported when trying to erase, rename or open a non-existent file.

3 Path not found

Reported by the directory handling routines when a path does not exist or is invalid. Also reported when trying to access a non-existent file.

4 Too many open files

The maximum number of files currently opened by your process has been reached. Certain operating systems limit the number of files which can be opened concurrently, and this error can occur when this limit has been reached.

5 File access denied

Permission to access the file is denied. This error might be caused by one of several reasons:

- Trying to open for writing a file which is read-only, or which is actually a directory.
- File is currently locked or used by another process.
- Trying to create a new file, or directory while a file or directory of the same name already exists.
- Trying to read from a file which was opened in write-only mode.
- Trying to write from a file which was opened in read-only mode.
- Trying to remove a directory or file while it is not possible.
- No permission to access the file or directory.

6 Invalid file handle

If this happens, the file variable you are using is trashed; it indicates that your memory is corrupted.

12 Invalid file access code

Reported when a reset or rewrite is called with an invalid FileMode value.

15 Invalid drive number

The number given to the GetDir or ChDir function specifies a non-existent disk.

16 Cannot remove current directory

Reported when trying to remove the currently active directory.

17 Cannot rename across drives

You cannot rename a file such that it would end up on another disk or partition.

100 Disk read error

An error occurred when reading from disk. Typically happens when you try to read past the end of a file.

101 Disk write error

Reported when the disk is full, and you're trying to write to it.

102 File not assigned

This is reported by Reset, Rewrite, Append, Rename and Erase, if you call them with an unassigned file as a parameter.

103 File not open

Reported by the following functions : Close, Read, Write, Seek, Eof, FilePos, FileSize, Flush, BlockRead, and BlockWrite if the file is not open.

104 File not open for input

Reported by Read, BlockRead, Eof, Eoln, SeekEof or SeekEoln if the file is not opened with Reset.

105 File not open for output

Reported by write if a text file isn't opened with Rewrite.

106 Invalid numeric format

Reported when a non-numeric value is read from a text file, and a numeric value was expected.

150 Disk is write-protected

(Critical error)

151 Bad drive request struct length

(Critical error)

152 Drive not ready

(Critical error)

154 CRC error in data

(Critical error)

156 Disk seek error

(Critical error)

157 Unknown media type

(Critical error)

158 Sector Not Found

(Critical error)

159 Printer out of paper

(Critical error)

160 Device write fault

(Critical error)

161 Device read fault

(Critical error)

162 Hardware failure

(Critical error)

200 Division by zero

The application attempted to divide a number by zero.

201 Range check error

If you compiled your program with range checking on, then you can get this error in the following cases:

1. An array was accessed with an index outside its declared range.
2. Trying to assign a value to a variable outside its range (for instance an enumerated type).

202 Stack overflow error

The stack has grown beyond its maximum size (in which case the size of local variables should be reduced to avoid this error), or the stack has become corrupt. This error is only reported when stack checking is enabled.

203 Heap overflow error

The heap has grown beyond its boundaries. This is caused when trying to allocate memory explicitly with New, GetMem or ReallocMem, or when a class or object instance is created and no memory is left. Please note that, by default, Free Pascal provides a growing heap, i.e. the heap will try to allocate more memory if needed. However, if the heap has reached the maximum size allowed by the operating system or hardware, then you will get this error.

204 Invalid pointer operation

You will get this in several cases:

- if you call Dispose or Freemem with an invalid pointer
- in case New or GetMem is called, and there is no more memory available. The behaviour in this case depends on the setting of ReturnNilIfGrowHeapFails. If it is True, then Nil is returned. if False, then runerror 204 is raised.

205 Floating point overflow

You are trying to use or produce real numbers that are too large.

206 Floating point underflow

You are trying to use or produce real numbers that are too small.

207 Invalid floating point operation

Can occur if you try to calculate the square root or logarithm of a negative number.

210 Object not initialized

When compiled with range checking on, a program will report this error if you call a virtual method without having called its object's constructor.

211 Call to abstract method

Your program tried to execute an abstract virtual method. Abstract methods should be overridden, and the overriding method should be called.

212 Stream registration error

This occurs when an invalid type is registered in the objects unit.

213 Collection index out of range

You are trying to access a collection item with an invalid index (objects unit).

214 Collection overflow error

The collection has reached its maximal size, and you are trying to add another element (objects unit).

215 Arithmetic overflow error

This error is reported when the result of an arithmetic operation is outside of its supported range. Contrary to Turbo Pascal, this error is only reported for 32-bit or 64-bit arithmetic overflows. This is due to the fact that everything is converted to 32-bit or 64-bit before doing the actual arithmetic operation.

216 General Protection fault

The application tried to access invalid memory space. This can be caused by several problems:

1. Dereferencing a nil pointer.
2. Trying to access memory which is out of bounds (for example, calling `move` with an invalid length).

217 Unhandled exception occurred

An exception occurred, and there was no exception handler present. The `sysutils` unit installs a default exception handler which catches all exceptions and exits gracefully.

218 Invalid value specified

Error 218 occurs when an invalid value was specified to a system call, for instance when specifying a negative value to a `seek()` call.

219 Invalid typecast

Thrown when an invalid typecast is attempted on a class using the `as` operator. This error is also thrown when an object or class is typecast to an invalid class or object and a virtual method of that class or object is called. This last error is only detected if the `-CR` compiler option is used.

222 Variant dispatch error

No dispatch method to call from variant.

223 Variant array create

The variant array creation failed. Usually when there is not enough memory.

224 Variant is not an array

This error occurs when a variant array operation is attempted on a variant which is not an array.

225 Var Array Bounds check error

This error occurs when a variant array index is out of bounds.

227 Assertion failed error

An assertion failed, and no `AssertErrorProc` procedural variable was installed.

229 Safecall error check

This error occurs if a safecall check fails, and no handler routine is available.

231 Exception stack corrupted

This error occurs when the exception object is retrieved and none is available.

232 Threads not supported

Thread management relies on a separate driver on some operating systems (notably, Unixes). The unit with this driver needs to be specified on the uses clause of the program, preferably as the first unit (`cthreads` on unix).